

# Probability & Statistics

## Unit-I Descriptive Statistics:

Statistic Introduction, population vs Sample, collection of data, primary and secondary data, measure of central tendency, measures of variability (spread or variance), Skewness, kurtosis, correlation, correlation coefficient, rank correlation, regression coefficients, method of least squares, regression lines.

## Unit-II Probability:

probability, probability axioms, addition law and multiplicative law of probability, conditional probability, Baye's Theorem, Random variables (distinct and continuous) probability density functions, properties, mathematical expectation.

## Unit-III Probability Distribution:

probability distributions, Binomial, poisson and Normal - their properties (Chebyshev's inequality), Approximation of the binomial distribution to normal distribution.

## Unit-IV Estimation and Testing of hypothesis

### Largest Sample Tests:

Estimation - parameters, statistics, sampling distribution, point estimations, Formulation of null hypothesis, alternative hypothesis, the crucial and acceptance regions, level of significance, two types of errors and power of the tests large sample Tests:

Test for single proportion, difference of proportions, test for single mean and difference of means. Confidence interval for parameters in one sample and two sample problems.

### Unit-V Small Sample Tests:

Student t-distribution (test for single mean, two means and paired t-test), testing of equality of variances (F-test),  $\chi^2$ -test for goodness of fit,  $\chi^2$ -test for independence of attributes.

# 1. Descriptive Statistics

## Topics:

→ Descriptive Statistics

→ measures of Central Tendency and measures of variability

→ Correlation

→ Regression.

**Correlation:** Relation between two variables is called correlation coefficient (or) coefficient of correlation.

**Covariance:** Let  $x$  &  $y$  be the two random variables with means  $\bar{x}$  and  $\bar{y}$  respectively. Then the covariance between two variables  $x$  and  $y$  defined by the relation  $\text{cov}(x, y) = E(xy) - E(x)E(y)$  ( $E = \text{Expectation \& Average}$ )

$$\bar{x} = \frac{\sum x}{n_1}, \quad \bar{y} = \frac{\sum y}{n_2}$$

## Note:

→ If  $x$  and  $y$  are independent Random variables

then  $\text{Cov}(x, y) = 0$  (or)  $\text{Cov}(x, y) = 0$

→ If  $a$  and  $b$  are constants then

$$* \text{Cov}(x+a, y+b) = \text{Cov}(x, y)$$

$$* \text{Cov}(ax, by) = ab \text{Cov}(x, y)$$

# Formulae: Correlation Coefficient:

$$r = \frac{\text{Cor}(x, y)}{\sigma_x \sigma_y}$$

$$r = \frac{N \sum xy - (\sum x)(\sum y)}{\sqrt{N \sum x^2 - (\sum x)^2} \sqrt{N \sum y^2 - (\sum y)^2}}$$

(or)

$$r(x, y) = p(x, y) = r = p = \frac{\sum xy}{\sqrt{\sum x^2 \sum y^2}}$$

1. Calculate coefficient of correlation from the following data

|   |    |   |   |    |    |    |   |
|---|----|---|---|----|----|----|---|
| x | 12 | 9 | 8 | 10 | 11 | 13 | 7 |
| y | 14 | 8 | 6 | 9  | 11 | 12 | 3 |

Given the data

The above values are small

| x  | y  | x <sup>2</sup> | y <sup>2</sup> | xy  |
|----|----|----------------|----------------|-----|
| 12 | 14 | 144            | 196            | 168 |
| 9  | 8  | 81             | 64             | 72  |
| 8  | 6  | 64             | 36             | 48  |
| 10 | 9  | 100            | 81             | 90  |
| 11 | 11 | 121            | 121            | 121 |
| 13 | 12 | 169            | 144            | 156 |
| 7  | 3  | 49             | 9              | 21  |

$$\sum x = 70 \quad \sum y = 63 \quad \sum x^2 = 728 \quad \sum y^2 = 651 \quad \sum xy = 676$$

We know that

$$r = \frac{N \sum xy - \sum x \sum y}{\sqrt{N \sum x^2 - (\sum x)^2} \sqrt{N \sum y^2 - (\sum y)^2}}$$

$$r = \frac{7(676) - (70)(63)}{\sqrt{7(728) - (70)^2} \sqrt{7(651) - (63)^2}}$$

$$r = \frac{4732 - 4410}{\sqrt{5096 - 4900} \sqrt{4557 - 3969}}$$

$$r = \frac{322}{\sqrt{196} \sqrt{588}}$$

$$r = \frac{322}{14(24.218)}$$

$$r = \frac{322}{339.4819}$$

$$r = \underline{\underline{0.95}}$$

2. Find the correlation coefficient of the following data

|   |    |    |    |    |    |    |    |    |    |    |
|---|----|----|----|----|----|----|----|----|----|----|
| x | 50 | 50 | 55 | 60 | 65 | 65 | 65 | 60 | 60 | 60 |
| y | 11 | 13 | 14 | 16 | 16 | 15 | 15 | 14 | 13 | 13 |

Given the data

The above values are big.

| x  | y  | $x = x - \bar{x}$ | $y = y - \bar{y}$ | $x^2$ | $y^2$ | $xy$ |
|----|----|-------------------|-------------------|-------|-------|------|
| 50 | 11 | -9                | -3                | 81    | 9     | 27   |
| 55 | 13 | -4                | -1                | 16    | 1     | 4    |
| 55 | 14 | -4                | 0                 | 16    | 0     | 0    |
| 60 | 16 | 1                 | 2                 | 1     | 4     | 2    |
| 65 | 16 | 6                 | 2                 | 36    | 4     | 12   |
| 65 | 15 | 6                 | 1                 | 36    | 1     | 6    |

|    |    |   |    |    |   |    |
|----|----|---|----|----|---|----|
| 65 | 15 | 6 | 1  | 36 | 1 | 36 |
| 60 | 14 | 1 | 0  | 1  | 0 | 0  |
| 60 | 13 | 1 | -1 | 1  | 1 | -1 |
| 60 | 13 | 1 | -1 | 1  | 1 | -1 |

$$\Sigma x = 590 \quad \Sigma y = 140 \quad \Sigma x^2 = 290 \quad \Sigma y^2 = 22 \quad \Sigma xy = 60$$

$$\bar{x} = \frac{\Sigma x}{n}, \quad \bar{y} = \frac{\Sigma y}{n}$$

$$\bar{x} = \frac{590}{10}, \quad \bar{y} = \frac{140}{10}$$

$$\bar{x} = 59, \quad \bar{y} = 14$$

We know that

$$r = \frac{N \Sigma xy - \Sigma x \Sigma y}{\sqrt{N \Sigma x^2 - (\Sigma x)^2} \sqrt{N \Sigma y^2 - (\Sigma y)^2}}$$

$$r = \frac{10(60) - (590)(140)}{\sqrt{10(290) - (590)^2} \sqrt{10(22) - (140)^2}}$$

$$r = \frac{600}{\sqrt{2900} \sqrt{220}}$$

$$r = \frac{600}{(53.851)(14.832)}$$

$$r = \frac{600}{798.71}$$

$$r = 0.7512$$

Q. Calculate the correlation coefficient for the following heights (in inches) of fathers (x) and

their sons (y)

|   |    |    |    |    |    |    |    |    |
|---|----|----|----|----|----|----|----|----|
| x | 65 | 66 | 67 | 67 | 68 | 69 | 70 | 72 |
| y | 67 | 68 | 65 | 68 | 72 | 72 | 69 | 71 |

Given The data

| x          | y          | $X = x - \bar{x}$ | $Y = y - \bar{y}$ | $X^2$     | $Y^2$     | $XY$      |
|------------|------------|-------------------|-------------------|-----------|-----------|-----------|
| 65         | 67         | -3                | -2                | 9         | 4         | 6         |
| 66         | 68         | -2                | -1                | 4         | 1         | 2         |
| 67         | 65         | -1                | -4                | 1         | 16        | 4         |
| 67         | 68         | -1                | -1                | 1         | 1         | 1         |
| 68         | 72         | 0                 | 3                 | 0         | 9         | 0         |
| 69         | 72         | 1                 | 3                 | 1         | 9         | 3         |
| 70         | 69         | 2                 | 0                 | 4         | 0         | 0         |
| 72         | 71         | 4                 | 2                 | 16        | 4         | 8         |
| <u>544</u> | <u>552</u> | <u>0</u>          | <u>0</u>          | <u>36</u> | <u>44</u> | <u>24</u> |

$$\bar{x} = \frac{\sum x}{n} \quad \bar{y} = \frac{\sum y}{n}$$

$$\bar{x} = \frac{544}{8} \quad \bar{y} = \frac{552}{8}$$

$$\bar{x} = 68 \quad \bar{y} = 69$$

We know that

$$r = \frac{N \sum XY - \sum X \sum Y}{\sqrt{N \sum X^2 - (\sum X)^2} \sqrt{N \sum Y^2 - (\sum Y)^2}}$$

$$r = \frac{8(24) - (0)(0)}{\sqrt{8(36) - (0)^2} \sqrt{8(44) - (0)^2}}$$

$$r = \frac{192}{\sqrt{288} \sqrt{352}}$$

$$r = \frac{192}{\sqrt{288} \sqrt{352}}$$

$$r = \frac{192}{(16.970)(18.76166)}$$

$$r = \frac{192}{318.3853}$$

$$r = 0.60304$$

$$\underline{\underline{r = 0.6030}}$$

### Rank Correlation Coefficient

Rank Correlation is applicable only to the individual observations. The formula for Spearman's rank correlation is given by

$$r = 1 - \frac{6 \sum D^2}{N(N^2 - 1)}$$

Where,

$r$  = Rank Coefficient of correlation

$D^2$  = Sum of the squares of the difference of two variables (or) ranks

$N$  = Number of paired observations

### Working Rules: I

→ When the ranks are given

1) Find the difference of two ranks and denoted by "D"

2. Square D and get  $D^2$
3. Obtain  $\rho$  by substituting the figures in the formula.

### Working Rule-II

→ When the ranks are not given, but actual data are given then we must give ranks we can give ranks by taking the highest as 1 or the lowest value as 1 next to the highest (lowest) as 2 and follow the same procedure for both the variables.

1. A Random Sample of Five college students is selected and their grades in mathematics and statistics are found to be

|             |    |    |    |    |    |
|-------------|----|----|----|----|----|
|             | 1  | 2  | 3  | 4  | 5  |
| mathematics | 85 | 60 | 73 | 40 | 90 |

|            |    |    |    |    |    |
|------------|----|----|----|----|----|
| statistics | 93 | 75 | 65 | 50 | 80 |
|------------|----|----|----|----|----|

Calculate Spearman's rank correlation coefficient

| marks in<br>mathematics<br>(x) | Ranks<br>(x) | marks in<br>statistics<br>(y) | Ranks<br>(y) | Rank<br>diff<br>$D = x - y$ | $D^2$ |
|--------------------------------|--------------|-------------------------------|--------------|-----------------------------|-------|
| 85                             | 2            | 93                            | 1            | 1                           | 1     |
| 60                             | 4            | 75                            | 3            | 1                           | 1     |
| 73                             | 3            | 65                            | 4            | -1                          | 1     |
| 40                             | 5            | 50                            | 5            | 0                           | 0     |
| 90                             | 1            | 80                            | 2            | -1                          | 1     |

$$\sum D^2 = 4$$

$N = 5$  (observations)

We know that

$$P = 1 - \frac{6 \sum D^2}{N(N^2 - 1)}$$

$$P = 1 - \frac{6(4)}{5(5^2 - 1)}$$

$$P = 1 - \frac{24}{5(24)}$$

$$P = 1 - \frac{1}{5}$$

$$P = 1 - \frac{1}{5}$$

$$P = \frac{4}{5}$$

$$\underline{\underline{P = 0.8}}$$

Q. The Ranks of 16 students in mathematics and statistics are as follows. (1, 1) (2, 10) (3, 3) (4, 4) (5, 5) (5, 3) (6, 7) (7, 2) (8, 6) (9, 8) (10, 11) (11, 15) (12, 9) (13, 14) (14, 12) (15, 16) (16, 13) Calculate the rank correlation coefficient for proficiencies of this group in mathematics and statistics.

Given the data

|   |   |    |   |   |   |   |   |   |   |    |    |    |    |    |
|---|---|----|---|---|---|---|---|---|---|----|----|----|----|----|
| x | 1 | 2  | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 |
| y | 1 | 10 | 3 | 4 | 3 | 5 | 7 | 2 | 6 | 8  | 11 | 15 | 9  | 14 |

|   |    |    |
|---|----|----|
| x | 15 | 16 |
| y | 16 | 13 |

| Ranks of<br>(x) | Ranks of<br>(y) | Ranks diff<br>$D = x - y$ | $D^2$ |
|-----------------|-----------------|---------------------------|-------|
| 1               | 1               | 0                         | 0     |
| 2               | 10              | -8                        | 64    |
| 3               | 3               | 0                         | 0     |
| 4               | 4               | 0                         | 0     |
| 5               | 3               | 2                         | 4     |
| 6               | 7               | -1                        | 1     |
| 7               | 2               | 5                         | 25    |
| 8               | 6               | 2                         | 4     |
| 9               | 8               | 1                         | 1     |
| 10              | 11              | -1                        | 1     |
| 11              | 15              | -4                        | 16    |
| 12              | 9               | 3                         | 9     |
| 13              | 14              | -1                        | 1     |
| 14              | 12              | 2                         | 4     |
| 15              | 14              | 1                         | 1     |
| 16              | 13              | 3                         | 9     |
|                 |                 |                           | <hr/> |
|                 |                 |                           | 140   |

$$N = 16$$

We know that

$$\begin{aligned}
 \rho &= 1 - \frac{6 \sum D^2}{N(N^2 - 1)} \\
 &= 1 - \frac{6(140)}{16(16^2 - 1)} \\
 &= 1 - \frac{840}{16(256 - 1)}
 \end{aligned}$$

$$= 1 - \frac{840}{16(255)}$$

$$= \frac{16(255) - 840}{16(255)}$$

$$= \frac{4080 - 840}{4080}$$

$$= \frac{3240}{4080}$$

$$= 0.79411$$

28/01

### Equal or Repeated Ranks

We know that the Rank correlation coefficient Formula

$$r = 1 - \frac{6 \left[ \sum D^2 + \frac{1}{12} (m^3 - m) + \frac{1}{12} (m^3 - m) + \dots \right]}{N^3 - N}$$

Where,  $m$  = the number of items whose ranks are common

1. Obtain The rank correlation coefficient for the following data.

|   |    |    |    |    |    |    |    |    |    |    |
|---|----|----|----|----|----|----|----|----|----|----|
| X | 68 | 64 | 75 | 50 | 64 | 80 | 75 | 40 | 55 | 64 |
| Y | 62 | 58 | 68 | 45 | 81 | 60 | 68 | 48 | 50 | 70 |

| $x$ | Ranks( $x$ ) | $y$ | Ranks( $y$ ) | $D$  | $D^2$    |
|-----|--------------|-----|--------------|------|----------|
| 68  | 4            | 62  | 5            | -1   | 1        |
| 64  | <del>6</del> | 58  | 7            | -1   | 1        |
| 75  | 2.5          | 68  | 3.5          | -1.0 | 1        |
| 50  | 9            | 45  | 10           | -1   | 1        |
| 64  | <del>6</del> | 81  | 1            | 5    | 25       |
| 80  | 1            | 60  | 6            | -5   | 25       |
| 75  | 2.5          | 68  | 3.5          | -1   | 1        |
| 40  | 10           | 48  | 9            | 1    | 1        |
| 55  | 8            | 50  | 8            | 0    | 0        |
| 64  | <del>6</del> | 70  | 2            | 4    | 16       |
|     |              |     |              |      | <hr/> 72 |

$$75 < \frac{2}{3} \quad \frac{2+3}{2} = \frac{5}{2} = 2.5 = m_1$$

$$64 < \frac{5}{6} \quad \frac{5+6+7}{3} = \frac{18}{3} = 6 = m_2$$

$$68 < \frac{3}{4} \quad \frac{3+4}{2} = \frac{7}{2} = 3.5 = m_3$$

We know that

$$\rho = 1 - \frac{6 \left[ \sum D^2 + \frac{1}{12} (m_1^3 - m_1) + \frac{1}{12} (m_2^3 - m_2) + \frac{1}{12} (m_3^3 - m_3) \right]}{(N^3 - N)}$$

$$= 1 - \frac{6 \left[ 72 + \frac{1}{12} (2^3 - 2) + \frac{1}{12} (3^3 - 3) + \frac{1}{12} (2^3 - 2) \right]}{(10^3 - 10)}$$

$$= 1 - \frac{6 \left[ 72 + \frac{1}{12} (8 - 2) + \frac{1}{12} (27 - 3) + \frac{1}{12} (8 - 2) \right]}{1000 - 10}$$

$$= 1 - \frac{6 \left( 72 + \frac{6}{12} + \frac{24}{12} + \frac{6}{12} \right)}{990}$$

$$= 1 - \frac{6 \left( 72 + \frac{1}{2} + 2 + \frac{1}{2} \right)}{990}$$

$$= 1 - \frac{6 \left( 74 + \frac{2}{2} \right)}{990}$$

$$= 1 - \frac{6(74+1)}{990}$$

$$= 1 - \frac{6(75)}{990}$$

$$= \frac{990 - (75)6}{990}$$

$$= \frac{915}{990}$$

$$= \frac{990 - 450}{990}$$

$$= \frac{540}{990} = \underline{\underline{0.5454}}$$

2 From the following data calculate the rank correlation coefficient

|   |    |    |    |   |    |    |    |    |    |    |
|---|----|----|----|---|----|----|----|----|----|----|
| x | 48 | 33 | 40 | 9 | 16 | 16 | 65 | 24 | 16 | 57 |
| y | 13 | 13 | 24 | 6 | 15 | 4  | 20 | 9  | 6  | 19 |

Ans  $\rho = 0.733$

| x  | Rank<br>(x) | y  | Rank<br>(y) | D    | D <sup>2</sup> |
|----|-------------|----|-------------|------|----------------|
| 48 | 3           | 13 | 5.5         | -1.5 | 2.25           |
| 33 | 5           | 13 | 5.5         | -0.5 | 0.25           |
| 40 | 4           | 24 | 1           | 3    | 9              |
| 9  | 10          | 6  | 8.5         | 1.5  | 2.25           |
| 16 | 8           | 15 | 4           | 4    | 16             |
| 16 | 8           | 4  | 10          | -2   | 4              |
| 65 | 1           | 20 | 2           | -1   | 1              |
| 24 | 6           | 9  | 7           | -1   | 1              |
| 16 | 8           | 6  | 8.5         | -0.5 | 0.25           |
| 57 | 2           | 19 | 3           | -1   | 1              |
|    |             |    |             |      | <hr/>          |

$$16 \begin{matrix} \swarrow 7 \\ 8 \\ \searrow 9 \end{matrix} \quad \frac{7+8+9}{3} = \frac{24}{3} = 8$$

~~44~~

$$13 \begin{matrix} \swarrow 5 \\ 6 \end{matrix} \quad \frac{5+6}{2} = \frac{11}{2} = 5.5$$

$$6 \begin{matrix} \swarrow 8 \\ 9 \end{matrix} \quad \frac{8+9}{2} = \frac{17}{2} = 8.5$$

We know that

$$\rho = 1 - \frac{6 \left[ \sum D^2 + \frac{1}{12} (m^3 - m) + \frac{1}{12} (m^3 - m) + \frac{1}{12} (m^3 - m) \right]}{N^3 - N}$$

$$= 1 - \frac{6 \left[ \cancel{44} + \frac{1}{2} (3^3 - 3) + \frac{1}{12} (2^3 - 2) + \frac{1}{12} (2^3 - 2) \right]}{10^3 - 10}$$

$$= 1 - \frac{6 \left[ \cancel{44} + \frac{1}{12} (24) + \frac{1}{12} (6) + \frac{1}{12} (6) \right]}{990}$$

990

$$= 1 - \frac{6 \left[ 41 + 2 + \frac{1}{2} + \frac{1}{2} \right]}{990}$$

$$= 1 - \frac{6 [41 + 2 + 1]}{990}$$

$$= 1 - \frac{6 [44]}{990}$$

$$= 1 - \frac{264}{990}$$

$$= \frac{990 - 264}{990}$$

$$= \frac{726}{990}$$

$$= \underline{\underline{0.733}}$$

c 29/01

### Regression:

Regression is a mathematical measure of the average relationship between two or more variables in terms of the original limits of the data.

Lines of Regression:

→ The line of regression of  $y$  on  $x$  is given by

$$y - \bar{y} = r \frac{\sigma_y}{\sigma_x} (x - \bar{x})$$

→ The line of regression of  $x$  on  $y$  is given by

$$x - \bar{x} = r \frac{\sigma_x}{\sigma_y} (y - \bar{y})$$

Regression coefficient:

→ Regression coefficient of  $y$  on  $x$  is  $b_{yx} = r \frac{\sigma_y}{\sigma_x}$

→ Regression Coefficients of  $x$  on  $y$  is  $b_{xy} = r \frac{\sigma_x}{\sigma_y}$

Correlation coefficient  $r = \pm \sqrt{b_{xy} b_{yx}}$

where  $b_{xy} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (y - \bar{y})^2}$

$$b_{yx} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

Statement: Correlation coefficient is the Geometric mean between the two regression coefficients

We know that  $r = \pm \sqrt{b_{xy} b_{yx}}$

$$r^2 = b_{xy} b_{yx}$$

where  $b_{xy} = r \frac{\sigma_x}{\sigma_y}$ ;  $b_{yx} = r \frac{\sigma_y}{\sigma_x}$

Angle between two regression lines

If  $\theta$  is the angle between two regression lines then  $\tan \theta = \left( \frac{1 - r^2}{r} \right) \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2}$

Regression Equation:

The standard form of regression equation is

$$y = a + bx \quad (\text{Regression coefficient of } y \text{ on } x)$$

$$x = a + by \quad (\text{Regression coefficient of } x \text{ on } y)$$

Regression Equation of  $y$  on  $x$ :

Normal equations are

$$\sum y = Na + b \sum x$$

$$\sum xy = a \sum x + b \sum x^2$$

## Regression Equation of $x$ on $y$ :

Normal equation are

$$\Sigma x = Na + b \Sigma y$$

$$\Sigma xy = a \Sigma y + b \Sigma y^2$$

Uses:

→ It is used to estimate the relation between two economic variables like income and expenditure.

→ It is a highly valuable tool in economics and business.

T. From the following data

- (i) The two regression equation
- (ii) The coefficient of correlation between the marks in Economics and statistics.
- (iii) The most likely marks in statistics when marks in Economics are 30.

marks in

Economics

$x$

25

28

35

32

31

36

29

38

34

marks in

Statistics

$y$

43

46

49

41

36

32

31

30

33

| $x$        | $y$        | $x - \bar{x}$ | $y - \bar{y}$ | $(x - \bar{x})(y - \bar{y})$ | $(x - \bar{x})^2$ | $(y - \bar{y})^2$ |
|------------|------------|---------------|---------------|------------------------------|-------------------|-------------------|
| 25         | 43         | -7            | 5             | -35                          | 49                | 25                |
| 28         | 46         | -4            | 8             | -32                          | 16                | 64                |
| 35         | 49         | 3             | 11            | 33                           | 9                 | 121               |
| 32         | 41         | 0             | 3             | 0                            | 0                 | 9                 |
| 31         | 36         | -1            | -2            | 2                            | 1                 | 4                 |
| 36         | 32         | 4             | -6            | -24                          | 16                | 36                |
| 29         | 31         | -3            | -7            | 21                           | 9                 | 49                |
| 38         | 30         | 6             | -8            | -48                          | 36                | 64                |
| 34         | 33         | 2             | -5            | -10                          | 4                 | 25                |
| 32         | 39         | 0             | 1             | 0                            | 0                 | 1                 |
| <u>320</u> | <u>380</u> | <u>-9</u>     | <u>-10</u>    | <u>-93</u>                   | <u>140</u>        | <u>398</u>        |

$$\bar{x} = \frac{\sum x}{n}, \quad \bar{y} = \frac{\sum y}{n}$$

$$\bar{x} = \frac{320}{10}, \quad \bar{y} = \frac{380}{10}$$

$$\bar{x} = 32, \quad \bar{y} = 38$$

$$b_{xy} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (y - \bar{y})^2} = \frac{-93}{398} = -0.23366$$

$$b_{yx} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2} = \frac{-93}{140} = -0.6643$$

(i) The line of regression of  $y$  on  $x$  is given by

$$y - \bar{y} = r \frac{\sigma_y}{\sigma_x} (x - \bar{x})$$

$$y - \bar{y} = b_{yx} (x - \bar{x})$$

$$y - 38 = -0.6643 (x - 32)$$

$$y = -0.6643(x-32) + 38$$

$$y = -0.6643x + 21.2576 + 38$$

$$y = -0.6643x + 59.2576 \rightarrow \textcircled{1}$$

The line of regression of  $x$  on  $y$  is given by

$$x - \bar{x} = b_{xy}(y - \bar{y})$$

$$x - 32 = -0.2337(y - 38)$$

$$x = -0.2337(y - 38) + 32$$

$$x = -0.2337y + 8.8806 + 32$$

$$x = -0.2337y + 39.4784 / 40.8806$$

$$x = -0.2337y + 40.8806 \rightarrow \textcircled{2}$$

(ii) The Coefficient of correlation

$$r = \pm \sqrt{b_{xy} b_{yx}}$$

$$r^2 = b_{xy} b_{yx}$$

$$r = \sqrt{(-0.2337)(-0.664)} \quad \cdot$$

$$r = \pm \sqrt{0.15517}$$

$$r = \pm 0.39391$$

$$r = \pm 0.3940$$

(iii) Given when  $x = 30$ ,  $y = ?$

$$y = -0.6643x + 59.2575$$

$$y = -0.6643(30) + 59.2575$$

$$y = -19.929 + 59.2575$$

$$\underline{y = 39.}$$

- d. The regression equations are  $3x + 2y = 26$  and  $6x + y = 31$ . Find the C.C between  $x$  and  $y$ . Also find means.

Since the equations passing through the points  $(\bar{x}, \bar{y})$   
put  $x = \bar{x}$ ,  $y = \bar{y}$

Given  $3\bar{x} + 2\bar{y} = 26 \rightarrow \textcircled{1}$

$$6\bar{x} + \bar{y} = 31 \rightarrow \textcircled{2}$$

From  $\textcircled{1} \times 1 \Rightarrow 3\bar{x} + 2\bar{y} = 26$

$$\begin{array}{r} \textcircled{2} \times 2 \Rightarrow 12\bar{x} + 2\bar{y} = 62 \\ \underline{(-) \quad (-) \quad (+)} \\ -9\bar{x} \quad = -36 \end{array}$$

$$\boxed{\bar{x} = 4}$$

$x = 4$  sub in  $\textcircled{1}$

From  $\textcircled{1} \Rightarrow 3(4) + 2y = 26$

$$12 + 2y = 26$$

$$2y = 14$$

$$\boxed{\bar{y} = 7}$$

means  $x = 4$ ,  $y = 7$

Correlation coefficient

From  $\textcircled{1} \Rightarrow 3x + 2y = 26$

$$3x = 26 - 2y$$

$$3x = -2y + 26 \quad (x \text{ on } y)$$

$$x = \frac{-2y}{3} + \frac{26}{3}$$

$$x = -\frac{2}{3}y + \frac{26}{3}$$

Here  $b_{xy} = -\frac{2}{3}$

From ②  $\Rightarrow 6x + y = 31 \quad (y \text{ on } x)$

$$6x = 31 - y$$

$$6x = -y + 31$$

$$y = 31 - 6x$$

$$y = -6x + 31$$

Here  $b_{yx} = -6$

Correlation Coefficient (WKT)

$$r = \pm \sqrt{b_{xy} b_{yx}}$$

$$r = \pm \sqrt{\left(-\frac{2}{3}\right)(-6)}$$

$$r = \pm \sqrt{\frac{12}{3}}$$

$$r = \pm \sqrt{4}$$

$$r = \pm 2 \quad [\because -1 \leq r \leq 1]$$

$$b_{xy} = -\frac{3}{2} \quad (y \text{ on } x)$$

$$b_{yx} = -\frac{1}{6} \quad (x \text{ on } y)$$

Correlation Coefficient  $= r^2 = b_{xy} b_{yx}$

$$r^2 = \left(-\frac{3}{2}\right)\left(-\frac{1}{6}\right)$$

$$r^2 = \left(\frac{3}{2}\right)\left(\frac{1}{6}\right)$$

$$r^2 = \frac{3}{12} = \frac{1}{4}$$

$$r^2 = \frac{1}{4}$$

$$r = \pm \sqrt{\frac{1}{4}}$$

$$r = \pm \frac{1}{2}$$

$$\underline{\underline{r = \pm 0.5}}$$

3. The equation of two regression lines are  $3x + 12y = 19$  &  $3x + 9y = 46$ . Find  $\bar{x}$  and  $\bar{y}$  and the correlation coefficient between  $x$  and  $y$ .

4. Calculate the correlation coefficient and regression coefficient for the following data.

|     |   |   |   |    |    |    |    |
|-----|---|---|---|----|----|----|----|
| $x$ | 2 | 4 | 6 | 8  | 10 | 12 | 14 |
| $y$ | 4 | 2 | 5 | 10 | 14 | 11 | 12 |

Find the estimate of  $y$  when  $x = 13$ .

5. The two lines of regression are  $8x - 10y + 66 = 0$  and  $40x - 18y - 214 = 0$ . The variance of  $x$  is 9.

Find (i) the mean values of  $x$  and  $y$

(ii) Correlation coefficient between  $x$  and  $y$ .

6. Determine the equation of a straight line which best fits the data

x 10 12 13 16 17 20 25

y 10 22 24 27 29 33 37

4) For the following data Find the equation of the two regression lines.

x 1 2 3 4 5

y 15 25 35 45 55

8. Find the most likely production corresponding to a rainfall 40 from the following data.

|                              | Rainfall (x) | production (y) |
|------------------------------|--------------|----------------|
| Average                      | 30           | 500 kgs        |
| S.D (Standard Deviation)     | 5            | 100 kgs        |
| CC (Correlation Coefficient) | 0.8          | -              |

We have to calculate the value of y when x  
So we have to find regression equation of y on x

We know that

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

Given data

$$\bar{x} = 30, \quad \bar{y} = 500$$

$$\bar{x} = 30, \bar{y} = 500$$

$$r = 0.8$$

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

$$y - 500 = b_{yx}(x - 30)$$

$$y - 500 = r \frac{\bar{y}}{\bar{x}} (x - 30)$$

$$y - 500 = (0.8) \left( \frac{500}{30} \right) (x - 30)$$

$$y - 500 = (0.8)(20)(x - 30)$$

$$y - 500 = 16(x - 30)$$

$$y = 16(x - 30) + 500$$

$$y = 16x - 480 + 500$$

$$y = 16x + 20$$

where  $x = 40$

$$y = 16(40) + 20$$

$$y = 640 + 20$$

$$y = \underline{660 \text{ kg.}}$$

9. Given  $n = 10$ ,  $\bar{x} = 5.41$ ,  $\bar{y} = 6.2$  and sum of the product of deviations from the mean of  $x$  and  $y$  is 66. Find the correlation coefficient

$$\sum (x - \bar{x})(y - \bar{y}) = 66$$

Given  $n = 10$ ,  $\bar{x} = 5.41$ ,  $\bar{y} = 6.2$ ,  $\sum (x - \bar{x})(y - \bar{y}) = 66$

To find, C.C

We know that

$$\rho = r = \frac{\sum xy}{\sum x^2 \sum y^2}$$

$$= \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}} \rightarrow \textcircled{1}$$

We know that

$$\sigma_x^2 = \frac{1}{n} \sum (x - \bar{x})^2$$

$$(5.4)^2 = \frac{1}{10} \sum (x - \bar{x})^2$$

$$10(5.4)^2 = \sum (x - \bar{x})^2$$

$$291.6 = \sum (x - \bar{x})^2$$

| x | y | x - $\bar{x}$ | y - $\bar{y}$ | (x - $\bar{x}$ ) <sup>2</sup> | (y - $\bar{y}$ ) <sup>2</sup> | (x - $\bar{x}$ )(y - $\bar{y}$ ) |
|---|---|---------------|---------------|-------------------------------|-------------------------------|----------------------------------|
|   |   |               |               |                               |                               |                                  |

We know that

$$\sigma_y^2 = \frac{1}{n} \sum (y - \bar{y})^2$$

$$(6.2)^2 = \frac{1}{10} \sum (y - \bar{y})^2$$

$$10(6.2)^2 = \sum (y - \bar{y})^2$$

$$384.4 = \sum (y - \bar{y})^2$$

From  $\textcircled{1} = \frac{66}{\sqrt{291.6 \times 384.4}}$

$$= \frac{66}{\sqrt{112091}}$$

$$= \frac{66}{334.79}$$

$$= \frac{66}{334.79}$$

$$= \underline{\underline{0.1971}}$$

3. Given two equation

$$3x + 12y = 19 \rightarrow ①$$

$$3x + 9y = 46 \rightarrow ②$$

Since the equations passing through the points

$(\bar{x}, \bar{y})$

From ① & ②

$$3\bar{x} + 12\bar{y} = 19 \rightarrow ①$$

$$3\bar{x} + 9\bar{y} = 46 \rightarrow ②$$

$$3\bar{y} = -27$$

$$3\bar{y} = -27$$

$$\bar{y} = \frac{-27}{3}$$

$$\boxed{\bar{y} = -9}$$

$\bar{y} = -9$  Sub in eq ①

$$3\bar{x} + 12(-9) = 19$$

$$3\bar{x} + (-108) = 19$$

$$3\bar{x} - 108 = 19$$

$$3\bar{x} = 19 + 108$$

$$\bar{x} = 127/3$$

$$\boxed{\bar{x} = 42.33}$$

means  $\bar{x} = 42.33, \bar{y} = -9$

Correlation coefficient

From ①  $3x + 12y = 19$

$$3x = 19 - 12y$$

$$3x = -12y + 19 \quad (x \text{ on } y)$$

$$x = \frac{-12y}{3} + \frac{19}{3}$$

$$x = -4y + \frac{19}{3}$$

$$\boxed{b_{xy} = -4}$$

From eq ②  $3x + 9y = 46 \quad (y \text{ on } x)$

$$9y = 46 - 3x$$

$$y = \frac{46}{9} - \frac{3}{9}x$$

$$y = -\frac{3}{9}x + \frac{46}{9}$$

$$y = -\frac{1}{3}x + \frac{46}{9}$$

$$\boxed{b_{yx} = -\frac{1}{3}}$$

Correlation coefficient

$$r = \pm \sqrt{b_{xy} b_{yx}}$$

$$r = \pm \sqrt{-4 \left(-\frac{1}{3}\right)}$$

$$r = \pm \sqrt{\frac{4}{3}}$$

$$r = \pm \underline{\underline{0.6666}}$$

A.

|   |   |   |   |    |    |    |    |
|---|---|---|---|----|----|----|----|
| x | 2 | 4 | 6 | 8  | 10 | 12 | 14 |
| y | 4 | 2 | 5 | 10 | 14 | 11 | 12 |

Given data above

| $x$       | $y$       | $x - \bar{x}$ | $y - \bar{y}$ | $(x - \bar{x})(y - \bar{y})$ | $(x - \bar{x})^2$ | $(y - \bar{y})^2$ |
|-----------|-----------|---------------|---------------|------------------------------|-------------------|-------------------|
| 2         | 4         | -6            | -4.29         | 25.74                        | 36                | 18.4              |
| 4         | 2         | -4            | -6.29         | 25.16                        | 16                | 39.5              |
| 6         | 5         | -2            | -3.29         | 6.58                         | 4                 | 43.3              |
| 8         | 10        | 0             | +1.71         | 0                            | 0                 | 2.3               |
| 10        | 14        | 2             | 5.71          | 11.42                        | 4                 | 32.6              |
| 12        | 11        | 4             | 2.71          | 10.84                        | 16                | 7.35              |
| 14        | 12        | 6             | 3.71          | 22.26                        | 36                | 13.8              |
| <u>56</u> | <u>58</u> | <u>0</u>      | <u>0</u>      | <u>102</u>                   | <u>112</u>        | <u>157</u>        |

$$\bar{x} = \frac{\sum x}{n} \quad \bar{y} = \frac{\sum y}{n}$$

$$\bar{x} = \frac{56}{7} \quad \bar{y} = \frac{58}{7}$$

$$\bar{x} = 8 \quad \bar{y} = 8.29$$

$$b_{xy} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (y - \bar{y})^2} = \frac{102}{157} = 0.65$$

$$b_{yx} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2} = \frac{102}{112} = 0.91$$

(i) The line of regression of  $y$  on  $x$  is given by

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

$$y - 8.29 = 0.91(x - 8)$$

$$y = 0.91(x - 8) + 8.29$$

$$y = 0.91x - 7.28 + 8.29$$

$$y = 0.91x + 1.01$$

$$y = 0.91x + 1 \rightarrow \text{①}$$

The line of regression  $x$  on  $y$  is given by

$$(x - \bar{x}) = b_{xy}(y - \bar{y})$$

$$(x - 8) = 0.65(y - 8.29)$$

$$x = 0.65(y - 8.29) + 8$$

$$x = 0.65y - 5.39 + 8$$

$$x = 0.65y + 2.61 \rightarrow \textcircled{2}$$

Correlation Coefficient

$$r = \pm \sqrt{b_{xy} b_{yx}}$$

$$r = \pm \sqrt{(0.65)(0.91)}$$

$$r = \pm \sqrt{0.5915}$$

$$r = \pm 0.769$$

Then Here  $x = 13$ ;  $y = ?$

$$y = 0.91x + 1$$

$$y = 0.91(13) + 1$$

$$y = 11.83 + 1$$

$$\underline{\underline{y = 12.83}}$$

5). Given data

|     |    |    |    |    |    |    |    |
|-----|----|----|----|----|----|----|----|
| $x$ | 10 | 12 | 13 | 16 | 17 | 20 | 25 |
| $y$ | 10 | 22 | 24 | 27 | 29 | 33 | 37 |

| $x$        | $y$        | $(x - \bar{x})$ | $(y - \bar{y})$ | $(x - \bar{x})(y - \bar{y})$ | $(x - \bar{x})^2$ | $(y - \bar{y})^2$ |
|------------|------------|-----------------|-----------------|------------------------------|-------------------|-------------------|
| 10         | 10         | -6.14           | -16             | 98.24                        | 37.7              | 256               |
| 12         | 22         | -4.14           | -4              | 16.56                        | 17.14             | 16                |
| 13         | 24         | -3.14           | -2              | 6.28                         | 9.9               | 4                 |
| 16         | 27         | -0.14           | 1               | 0.14                         | 0.019             | 1                 |
| 17         | 29         | 0.86            | 3               | 2.58                         | 0.516             | 9                 |
| 20         | 33         | 3.86            | 7               | 27.02                        | 14.9              | 49                |
| 25         | 37         | 8.86            | 11              | 97.46                        | 78.5              | 121               |
| <u>113</u> | <u>182</u> | <u>0</u>        | <u>0</u>        | <u>248.28</u>                | <u>158.7</u>      | <u>456</u>        |

$$\bar{x} = \frac{\sum x}{n} \quad \bar{y} = \frac{\sum y}{n}$$

$$\bar{x} = \frac{113}{7} \quad \bar{y} = \frac{182}{7}$$

$$\bar{x} = 16.14 \quad \bar{y} = 26$$

$$b_{yx} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2} = \frac{248.28}{158.7} = 1.564$$

The equation of a straight line

$$(y - \bar{y}) = b_{yx}(x - \bar{x})$$

$$y - 26 = b_{yx}(x - 16.14)$$

$$y - 26 = 1.564(x - 16.14)$$

$$y = 1.564x - 25.24 + 26$$

$$y = \underline{1.564x - 51.24}$$

4. Given data

$$x: 1 \quad 2 \quad 3 \quad 4 \quad 5$$

$$y: 4 \quad 15 \quad 25 \quad 35 \quad 45 \quad 55$$

| $x$       | $y$        | $(x-\bar{x})$ | $(y-\bar{y})$ | $(x-\bar{x})(y-\bar{y})$ | $(x-\bar{x})^2$ | $(y-\bar{y})^2$ |
|-----------|------------|---------------|---------------|--------------------------|-----------------|-----------------|
| 1         | 15         | -2            | -20           | 40                       | 4               | 400             |
| 2         | 25         | -1            | -10           | 10                       | 1               | 100             |
| 3         | 35         | 0             | 0             | 0                        | 0               | 0               |
| 4         | 45         | 1             | 10            | 10                       | 1               | 100             |
| 5         | 55         | 2             | 20            | 40                       | 4               | 400             |
| <u>15</u> | <u>175</u> | <u>0</u>      | <u>0</u>      | <u>100</u>               | <u>10</u>       | <u>1000</u>     |

$$\bar{x} = \frac{\sum x}{n} = \frac{15}{5} = 3$$

$$\bar{y} = \frac{\sum y}{n} = \frac{175}{5} = 35$$

The line of regression  $y$  on  $x$  is

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

$$b_{yx} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2} = \frac{100}{10} = 10$$

$$y - \bar{y} = 10(x - \bar{x})$$

$$y - 35 = 10(x - 3)$$

$$y = 10x - 30 + 35$$

$$\underline{y = 10x + 5}$$

The line of regression  $x$  on  $y$  is

$$(x - \bar{x}) = b_{xy}(y - \bar{y})$$

$$b_{xy} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (y - \bar{y})^2} = \frac{100}{1000} = 0.1$$

$$x - 3 = 0.1(y - 35)$$

$$x = 0.1y - 3.5 + 3$$

$$x = \underline{\underline{0.1y - 0.5}}$$

5. Given equations

$$8x - 10y + 66 = 0 \Rightarrow 8x - 10y = -66 \rightarrow \textcircled{1}$$

$$40x - 18y - 214 = 0 \Rightarrow 40x - 18y = 214 \rightarrow \textcircled{2}$$

Since the two equations passing through the  $\bar{x}, \bar{y}$

From  $\textcircled{1}$  &  $\textcircled{2}$

$$8\bar{x} - 10\bar{y} = -66 \Rightarrow \times 5 \Rightarrow 40\bar{x} - 50\bar{y} = -330 \rightarrow \textcircled{1}$$

$$40\bar{x} - 18\bar{y} = 214$$

$$40\bar{x} - 18\bar{y} = 214 \rightarrow \textcircled{2}$$

$$+ 32\bar{y} = 1544$$

$$32\bar{y} = 1544$$

$$\bar{y} = \frac{1544}{32}$$

$$\boxed{\bar{y} = 17}$$

$\boxed{\bar{y} = 17}$  Sub in eq  $\textcircled{2}$

$$40\bar{x} - 18(17) = 214$$

$$40\bar{x} - 306 = 214$$

$$40\bar{x} = 214 + 306$$

$$40\bar{x} = 520$$

$$\bar{x} = \frac{520}{40}$$

$$\boxed{\bar{x} = 13}$$

(i) means  $\bar{x} = 13, \bar{y} = 17$

(ii) Correlation Coefficient

$$\text{From eq } \textcircled{1} \Rightarrow 8x - 10y = -66$$

$$8x = -66 + 10y$$

$$8x = 10y - 66$$

$$x = \frac{10^5}{84} y - \frac{6.6}{8} \quad \# (x \text{ on } y)$$

$$x = \frac{5}{4} y - \frac{66}{8}$$

$$\boxed{b_{xy} = \frac{5}{4}} \quad \boxed{b_{xy} = \frac{4}{5}} \quad (y \text{ on } x)$$

$$\text{From (2)} \Rightarrow 40x - 18y = 214 \quad (y \text{ on } x)$$

$$-18y = 214 - 40x$$

$$-18y = -40x + 214$$

$$y = \frac{-40}{-18} x + \frac{214}{-18}$$

$$y = 2.22 \cdot x - \frac{214}{18}$$

$$\boxed{b_{yx} = 2.22} \quad \boxed{b_{yx} = \frac{18}{40}} \quad (x \text{ on } y)$$

Correlation coefficient

$$r = \pm \sqrt{b_{xy} b_{yx}}$$

$$r = \pm \sqrt{\frac{4}{5} \times \frac{18}{40}}$$

$$r = \pm \sqrt{(0.8)(0.45)}$$

$$r = \pm \sqrt{0.36}$$

$$\underline{\underline{r = \pm 0.6}}$$

Angle between two regression lines

We know that

$$\tan \theta = \left( \frac{1-r^2}{r} \right) \frac{\sigma_x \sigma_y}{\sigma_x^2 \sigma_y^2}$$

1. If  $\theta$  is an angle between two regression lines and standard deviation of  $y$  is twice the standard deviation of  $x$  and  $r=0.25$  find  $\tan \theta$ .

Given data,  $r=0.25$ ,  $\sigma_y = 2\sigma_x$

We know that

$$\tan \theta = \frac{1-r^2}{r} \frac{\sigma_x \sigma_y}{\sigma_x^2 \sigma_y^2}$$

$$\tan \theta = \frac{1-(0.25)^2}{0.25} \frac{\sigma_x 2\sigma_x}{\sigma_x^2 2\sigma_x^2}$$

$$\tan \theta = \frac{1-0.0625}{0.25} \frac{2\cancel{\sigma_x^2}}{2\cancel{\sigma_x^2}}$$

$$\tan \theta = \frac{0.9375}{0.25} \left( \frac{2}{2} \right)$$

$$\tan \theta = 3.75 \times \frac{2}{2}$$

$$\tan \theta = \underline{\underline{1.5}}$$

# Measure of Central Tendency and Variability

Some important measures of central tendency

→ Arithmetic mean

→ median

→ mode

→ Geometric mean

→ Harmonic mean

## Arithmetic mean:

→ Arithmetic mean of an ungrouped data

Let  $x_1, x_2, \dots, x_n$  be the  $n$  values of a variable,

then the arithmetic mean is given by

$$\text{Arithmetic mean} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{\sum x_i}{n}$$

→ Arithmetic mean of grouped data

Let  $x_1, x_2, \dots, x_n$  be the  $n$  classes of a frequency distribution with Frequency  $f_1, f_2, \dots, f_n$

$$\text{Arithmetic mean} = \boxed{\frac{\sum f_i x_i}{\sum f_i} = \frac{\sum f_i x_i}{N}}$$

Where  $N = \text{Total Frequency} = \sum f_i$

Note: To find arithmetic means

→ Direct method =  $\boxed{\bar{x} = \frac{\sum fx}{\sum f}}$

→ assumed mean method

$$\boxed{\bar{x} = a + \frac{\sum fd}{\sum f}}$$

→ step deviation method

$$\bar{x} = A + \frac{\Sigma fd}{\Sigma f} \times C$$

## Median

Working rules:

→ The given data we have to write ascending order or descending order

→ Median of an ungrouped data

\* If  $n$  is odd, then the median is the average values of  $\left(\frac{n+1}{2}\right)^{\text{th}}$  variate

\* If  $n$  is even, then the median is the average value of  $\left(\frac{n}{2}\right)^{\text{th}}$  variate and  $\left(\frac{n}{2} + 1\right)^{\text{th}}$  variate

i.e., 
$$\frac{\left(\frac{n}{2}\right)^{\text{th}} + \left(\frac{n}{2} + 1\right)^{\text{th}} \text{ term}}{2}$$

→ Median of a grouped data

We know that

$$\text{Median} = l + \frac{N/2 - m}{f} \times C$$

Where  $l$  = lower limit of the median class

$m$  = cumulative (add) Frequency of the class preceding the median class

$N$  = Total Frequency.

Mode: The value of the variate which Frequency maximum.

→ Mode of a grouped data

$$\text{mode} = l + \frac{f - f_1}{2f - f_1 - f_2} \times C$$

Where  $l$  = lower limit of the modal class

$f$  = Frequency of the modal class

$f_1$  = Frequency of the classes preceeding the modal class

$f_2$  = Frequency of the classes Succeeding the modal class

$C$  = width of modal class

Geometric mean:

→ Geometric mean of the ungrouped data

We know that

$$G = \text{Antilog} \left[ \frac{1}{N} \sum_{i=1}^N \log x_i \right]$$

→ Geometric mean of the grouped data

$$G = \text{Antilog} \left[ \frac{\sum f \log m}{N} \right]$$

Where  $m$  is the mid (point) value of the class interval

Harmonic mean:

→ Harmonic mean of an ungrouped data

$$H.m = \frac{n}{\sum_{i=1}^n \frac{1}{x_i}}$$

→ Harmonic mean of a grouped data

$$H.m = \frac{f_1 + f_2 + \dots + f_n}{\frac{f_1}{x_1} + \frac{f_2}{x_2} + \dots + \frac{f_n}{x_n}}$$

$$H.m = \frac{N}{\sum_{i=1}^n \frac{f_i}{x_i}} \quad [N = \sum f_i]$$

1. Calculate arithmetic mean of the following data 40, 50, 55, 78, 58, 60, 73, 35, 43, 48

$$\text{Arithmetic mean} = \bar{x} = \frac{\sum x_i}{n}$$

Given data,

40, 50, 55, 78, 58, 60, 73, 35, 43, 48

$$\sum x_i = 40 + 50 + 55 + 78 + 58 + 60 + 73 + 35 + 43 + 48$$

$$\sum x_i = 540$$

$$n = 10$$

$$\text{Arithmetic mean } \bar{x} = \frac{\sum x_i}{n} = \frac{540}{10} = \underline{\underline{54}}$$

2. From the following data find the mean profits

| Profit per shop | 100 - 200 | 200 - 300 | 300 - 400 | 400 - 500 | 500 - 600 | 600 - 700 | 700 - 800 |
|-----------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| No. of shops    | 10        | 18        | 20        | 26        | 30        | 28        | 18        |

| Sol | profit per shops (Rs) | mid value (m) | d = m - 450 | f  | fd    |
|-----|-----------------------|---------------|-------------|----|-------|
|     | (x)                   |               |             |    |       |
|     | 100-200               | 150           | -300        | 10 | -3000 |
|     | 200-300               | 250           | -200        | 18 | -3600 |
|     | 300-400               | 350           | -100        | 20 | -2000 |
|     | 400-500               | 450           | 0           | 26 | 0     |
|     | 500-600               | 550           | 100         | 30 | 3000  |
|     | 600-700               | 650           | 200         | 28 | 5600  |
|     | 700-800               | 750           | 300         | 18 | 5400  |
|     |                       |               | 150         |    | 5400  |

We know that we use Assumed mean method

$$\bar{x} = A + \frac{\sum fd}{N}$$

$$= 450 + \frac{\sum fd}{N}$$

$$= 450 + \frac{5400}{150}$$

$$= 450 + 36$$

$$= \underline{\underline{486}}$$

∴ Average profit is 486

3. Find the median From the following data  
57, 58, 61, 42, 38, 65, 72, 66

Given data

57, 58, 61, 42, 38, 65, 72, 66

The given data, we have to arrange ascending

Order : 38, 42, 57, 58, 61, 65, 66, 72  
① ② ③ ④ ⑤ ⑥ ⑦ ⑧

if 'n' is even number

$$\text{median} = \frac{(\frac{n}{2}) + (\frac{n}{2} + 1)}{2} \text{ term}$$

$$= \frac{\frac{8}{2} + (\frac{8}{2} + 1)}{2} \text{ term}$$

$$= \frac{4 + (4 + 1)}{2} \text{ term}$$

$$= \frac{4 + 5}{2} \text{ term}$$

$$= \frac{9}{2}$$

$$= 4.5$$

$$= \frac{58 + 61}{2} = \frac{119}{2} = \underline{\underline{59.5}}$$

4. Calculate the median from the following data

marks. 10-25 25-40 40-55 55-70 70-85 85-100

Frequency 6 20 44 26 3 1

We know that

$$\text{median} = l + \frac{N/2 - m}{f} \times c$$

Where

$l$  = lower limit of median class

$m$  = Cumulative frequency of the classes  
Preceding the median classes

$N$  = Total Frequency.

| marks  | Frequency | Cumulative Frequency |
|--------|-----------|----------------------|
| 10-25  | 6         | 6                    |
| 25-40  | 20        | 26                   |
| 40-55  | 44        | 70                   |
| 55-70  | 26        | 96                   |
| 70-85  | 3         | 99                   |
| 85-100 | 1         | 100                  |

Here  $N=100$ ,  $N/2 = 100/2 = 50$

Hence the class just greater than  $n/2$  is 40-55  
median class is 40-55

$$\text{median} = l + \frac{N/2 - m}{f} \times c \rightarrow (\text{class interval difference})$$

$$= 40 + \frac{50 - 26}{44} \times 15$$

$$= 40 + \frac{24}{44} \times 15$$

$$= 40 + 0.5454 \times 15$$

$$= 40 + 8.18$$

$$= \underline{\underline{48.18}}$$

5. Find the mode of the following data.

(i) 850, 750, 600, 825, 850, 725, 600, 850, 640, 8

(ii) 40, 45, 48, 57, 78

(iii) 45, 55, 50, 45, 40, 55, 45, 55

(i) mode = 850 (850 - 3 times repeated)

(ii) mode = 0 (No repeated values)

(iii) mode = 45 (45 repeated 3 times)

6. Find the mode of the following distribution

| Class Interval | 0-10 | 10-20 | 20-30 | 30-40 | 40-50 | 50-60 |
|----------------|------|-------|-------|-------|-------|-------|
| Frequency      | 5    | 8     | 7     | 12    | 28    | 20    |

We know that

| Class Interval | 60-70 | 70-80 | 80-90 |
|----------------|-------|-------|-------|
| Frequency      | 10    | 10    | 10    |

$$\text{mode} = l + \frac{f - f_1}{2f - f_1 - f_2} \times c$$

Where  $l$  = lower limit of the modal class (40)

$f$  = Frequency of the modal class (28)

$f_1$  = Frequency of the classes preceeding the modal class (12)

$f_2$  = Frequency of the classes succeeding the modal class (20)

$c$  = width of the modal class (10)

$$\text{mode} = l + \frac{f - f_1}{2f - f_1 - f_2} \times c$$

$$= 40 + \frac{28 - 12}{2(28) - 12 - 20} \times 10$$

$$= 40 + \frac{16}{56 - 12 - 20} \times 10$$

$$= 40 + \frac{16}{24} \times 10$$

$$= 40 + \frac{16}{24} \times 10$$

$$= 40 + 0.66 \times 10$$

$$= 40 + 6.66$$

$$= \underline{\underline{46.66}}$$

7. Calculate the Harmonic mean of the following data

|                   |   |   |   |   |    |    |
|-------------------|---|---|---|---|----|----|
| Size of the items | 6 | 7 | 8 | 9 | 10 | 11 |
| Frequency         | 4 | 6 | 9 | 5 | 2  | 8  |

| Size of the items<br>( $x$ ) | Frequency | $1/x$  | $f(1/x)$ |
|------------------------------|-----------|--------|----------|
| 6                            | 4         | 0.1667 | 0.6668   |
| 7                            | 6         | 0.1429 | 0.8574   |
| 8                            | 9         | 0.1250 | 1.1250   |
| 9                            | 5         | 0.1111 | 0.5555   |
| 10                           | 2         | 0.1000 | 0.2000   |
| 11                           | 8         | 0.0909 | 0.7292   |

$$\sum f_i = 34$$

$$\sum (1/x) = 4.13$$

We know that, the given data is ungrouped data

$$\begin{aligned} \text{Harmonic mean} &= \frac{n}{\sum \frac{1}{x_i}} \\ &= \frac{34}{4.13} \\ &= \underline{\underline{8.23}} \end{aligned}$$

8. Calculate the Harmonic mean of the following data

|           |       |       |       |       |       |       |    |
|-----------|-------|-------|-------|-------|-------|-------|----|
| marks     | 30-40 | 40-50 | 50-60 | 60-70 | 70-80 | 80-90 | 90 |
| Frequency | 15    | 13    | 8     | 6     | 15    | 7     | 6  |

| marks  | Frequency<br>(f)                  | mid value<br>(m) | $\frac{1}{m}$ | $fx(\frac{1}{m})$                                   |
|--------|-----------------------------------|------------------|---------------|-----------------------------------------------------|
| 30-40  | 15                                | 35               | 0.02859       | 0.42855                                             |
| 40-50  | 13                                | 45               | 0.02222       | 0.2886                                              |
| 50-60  | 8                                 | 55               | 0.01818       | 0.14544                                             |
| 60-70  | 6                                 | 65               | 0.01538       | 0.09204                                             |
| 70-80  | 15                                | 75               | 0.01333       | 0.19995                                             |
| 80-90  | 7                                 | 85               | 0.01176       | 0.08232                                             |
| 90-100 | 6                                 | 95               | 0.01053       | 0.06318                                             |
|        | <u><math>\Sigma f = 70</math></u> |                  |               | <u><math>\Sigma f(\frac{1}{m}) = 1.30034</math></u> |

We know that, the given data is grouped data

$$\text{Harmonic mean} = \frac{N}{\Sigma f(\frac{1}{m})}$$

$$= \frac{70}{1.30034} = \underline{53.83}$$

## Measures of Dispersion

The measures of Dispersion in common use are

- Range
- mean deviation
- Standard deviation

\* Range: maximum value - minimum value

\* mean deviation from mean of an ungrouped  
of data

$$\text{mean deviation} = \frac{\Sigma |x - x_0|}{n}$$

→ mean deviation for a grouped data

\* Discrete Frequency distribution

\* Continuous Frequency distribution

1) mean deviation about mean of a grouped data with discrete frequency distribution (DFD)

We know that,

$$\text{mean deviation} = \frac{\sum f_i |x_i - \bar{x}|}{\sum f_i} \quad \text{Where } \bar{x} = \frac{\sum x_i f_i}{\sum f_i}$$

2) mean deviation about median of a grouped data

We know that,

$$\text{mean deviation from median} = \frac{\sum f_i |x_i - \text{median}|}{\sum f_i}$$

1. Find the mean deviation about the mean for the following data.

|       |   |   |    |   |    |    |
|-------|---|---|----|---|----|----|
| $x_i$ | 2 | 5 | 7  | 8 | 10 | 35 |
| $f_i$ | 6 | 8 | 10 | 6 | 8  | 2  |

To Find mean deviation about the mean

We know that

$$\text{mean deviation} = \frac{\sum f_i |x_i - \bar{x}|}{\sum f_i}$$

$$\text{Where } \bar{x} = \frac{\sum x_i f_i}{\sum f_i}$$

| $x_i$ | $f_i$          | $x_i f_i$        | $(x_i - \bar{x})$ | $f_i (x_i - \bar{x})$ |
|-------|----------------|------------------|-------------------|-----------------------|
| 2     | 6              | 12               | 6                 | 36                    |
| 5     | 8              | 40               | 3                 | 24                    |
| 7     | 10             | 70               | 1                 | 10                    |
| 8     | 6              | 48               | 0                 | 0                     |
| 10    | 8              | 80               | 2                 | 16                    |
| 35    | $\frac{2}{40}$ | $\frac{70}{320}$ | 27                | $\frac{54}{140}$      |

$$\left\{ \begin{array}{l} \bar{x} = \frac{320}{40} \\ \bar{x} = 8 \end{array} \right.$$

We know that,

$$\begin{aligned} \text{Mean deviation} &= \frac{\sum f_i (x_i - \bar{x})}{\sum f_i} \\ &= \frac{140}{40} = \underline{\underline{3.5}} \end{aligned}$$

2. Calculate the mean deviation about median from the following data.

Size of items    6    7    8    9    10    11    12

Frequency        3    6    9    13    8    5    4

We know that the mean deviation from the

$$\text{median} = \frac{\sum f_i |x_i - \text{median}|}{\sum f_i}$$

Here no. of values  $n = 7$  (odd)

We know that  $(n+1)^{th}$

$$\text{median} = \frac{\left(\frac{n+1}{2}\right)^{th} \text{ term} + 1}{2} = \frac{9+1}{2} = \frac{10}{2} = 5^{th} \text{ term} = 9$$

| $x_i$ | $f_i$     | Cumulative Frequency | $ x_i - \text{median} $ | $f_i( x_i - \text{median} )$ |
|-------|-----------|----------------------|-------------------------|------------------------------|
| 6     | 3         | 3                    | 3                       | 9                            |
| 7     | 6         | 9                    | 2                       | 12                           |
| 8     | 9         | 18                   | 1                       | 9                            |
| 9     | 13        | 31                   | 0                       | 0                            |
| 10    | 8         | 39                   | 1                       | 8                            |
| 11    | 5         | 44                   | 2                       | 10                           |
| 12    | 4         | 48                   | 3                       | 12                           |
|       | <u>48</u> |                      |                         | <u>60</u>                    |

We know that

$$\text{mean deviation about median} = \frac{\sum f_i |x_i - \text{median}|}{\sum f_i}$$

$$= \frac{60}{48} = 1.25$$

19/02

Variance and standard deviation of Grouped / Ungrouped data

| Statistics     | Direct method                                                                                                                                  | Actual mean method                                                                                                                                                                     | Assumed mean method                                                                                                                                                    | Step deviation method.                                                                                     |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| ungrouped data | $\sigma = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$ or<br>$\sigma^2 = \frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2$ | $\sigma = \sqrt{\frac{\sum d^2}{n} - \frac{\left(\frac{\sum d}{n}\right)^2}{n}}$ $d = x - \bar{x}$ (or)<br>$\sigma^2 = \frac{\sum d^2}{n} - \frac{\left(\frac{\sum d}{n}\right)^2}{n}$ | $\sigma = \sqrt{\frac{\sum d^2}{n} - \left(\frac{\sum d}{n}\right)^2}$ $d = x - A$ (or)<br>$\sigma^2 = \frac{\sum d^2}{n} - \frac{\left(\frac{\sum d}{n}\right)^2}{n}$ | $\sigma = \sqrt{\frac{\sum d^2}{n} - \left(\frac{\sum d}{n}\right)^2}$ $d = x - \frac{A}{c}$               |
| grouped data   | —                                                                                                                                              | $\sigma = \sqrt{\frac{\sum f d^2}{N} - \frac{\left(\frac{\sum f d}{N}\right)^2}{N}}$ $d = x - \bar{x}$ $d^2 = (x - \bar{x})^2$                                                         | $\sigma = \sqrt{\frac{\sum f d^2}{N} - \frac{\left(\frac{\sum f d}{N}\right)^2}{N}}$ $d = (x - \bar{x})$                                                               | $\sigma = \sqrt{\frac{\sum f d^2}{N} - \frac{\left(\frac{\sum f d}{N}\right)^2}{N}}$ $d = x - \frac{A}{c}$ |

Standard deviation =  $\sigma = \sqrt{\text{variance}}$

- Find the variance and standard deviation of the following data

5, 12, 3, 18, 6, 8, 2, 10

| <u>x</u>                          | <u>x<sup>2</sup></u>                 | <u>(x - <math>\bar{x}</math>)</u> | <u>d</u> | <u>d<sup>2</sup></u>                 |
|-----------------------------------|--------------------------------------|-----------------------------------|----------|--------------------------------------|
| 5                                 | 25                                   | -3                                | -3       | 9                                    |
| 12                                | 144                                  | 4                                 | 4        | 16                                   |
| 3                                 | 9                                    | -5                                | -5       | 25                                   |
| 18                                | 324                                  | 10                                | 10       | 100                                  |
| 6                                 | 36                                   | -2                                | -2       | 4                                    |
| 8                                 | 64                                   | 0                                 | 0        | 0                                    |
| 2                                 | 4                                    | -6                                | -6       | 36                                   |
| 10                                | 100                                  | 2                                 | 2        | 4                                    |
| <u><math>\Sigma x = 64</math></u> | <u><math>\Sigma x^2 = 706</math></u> |                                   |          | <u><math>\Sigma d^2 = 194</math></u> |

$$\bar{x} = \frac{\Sigma x}{n} = \frac{64}{8} = 8$$

$$\begin{aligned}
 \text{Direct method } \sigma &= \sqrt{\frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2} \\
 &= \sqrt{\frac{706}{8} - 8^2} \\
 &= \sqrt{\frac{706}{8} - 64} \\
 &= \sqrt{88.25 - 64} \\
 &= \sqrt{24.25} = \underline{\underline{4.9244}}
 \end{aligned}$$

Assumed mean method  $\sigma = \sqrt{\frac{\Sigma d^2}{n} - \left(\frac{\Sigma d}{n}\right)^2}$

Actual

$$d = x - \bar{x}$$

$$\sigma = \sqrt{\frac{194}{8} + \dots}$$

$$\sigma = \sqrt{24.25}$$

$$\sigma = \underline{\underline{4.924}}$$

# Variance & Standard deviation for a Discrete Frequency Distribution

1. Calculate the standard deviation (or) mean standard deviation for the following distribution

$x_i$  - 4 8 11 17 20 24 32

$f_i$  3 5 9 5 4 3 1

| $x_i$      | $f_i$     | $d = x - \bar{x}$ | $f_i x_i$  | $d^2$ | $f d^2$     |
|------------|-----------|-------------------|------------|-------|-------------|
| 4          | 3         | -10               | 12         | 100   | 300         |
| 8          | 5         | -6                | 40         | 36    | 180         |
| 11         | 9         | -3                | 99         | 9     | 81          |
| 17         | 5         | +3                | 85         | 9     | 45          |
| 20         | 4         | 6                 | 80         | 36    | 144         |
| 24         | 3         | 4                 | 72         | 16    | 48          |
| 32         | 1         | 18                | 32         | 324   | 324         |
| <u>116</u> | <u>30</u> |                   | <u>420</u> |       | <u>1374</u> |

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{420}{30} = \underline{\underline{14}}$$

$$\begin{aligned} \text{Actual mean method } \sigma &= \sqrt{\frac{\sum f d^2}{\sum f}} \\ &= \sqrt{\frac{1374}{30}} \\ &= \sqrt{45.8} \\ &= \underline{\underline{6.77}} \end{aligned}$$

Q. Find the mean standard deviation of the following distribution

$x_i$  : 6 10 14 18 24 28 30

$f_i$  : 2 4 7 12 8 4 3

6.59

| $x_i$ | $f_i$     | $f_i x_i$  | $d = x - \bar{x}$ | $d^2$       | $f_i d^2$   |
|-------|-----------|------------|-------------------|-------------|-------------|
| 6     | 2         | 12         | -13               | 169         | 338         |
| 10    | 4         | 40         | -9                | 81          | 324         |
| 14    | 7         | 98         | -5                | 25          | 175         |
| 18    | 12        | 216        | -1                | 1           | 12          |
| 24    | 8         | 192        | 5                 | 25          | 200         |
| 28    | 4         | 112        | 9                 | 81          | 324         |
| 30    | 3         | 90         | 11                | 121         | 363         |
|       | <u>40</u> | <u>760</u> |                   | <u>1860</u> | <u>1736</u> |

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{760}{40} = 19$$

Actual mean method  $\sigma = \sqrt{\frac{\sum f_i d^2}{\sum f_i}}$

$$\sigma = \sqrt{\frac{1860}{40} \cdot \frac{1736}{40}}$$

$$\sigma = \sqrt{46.625}$$

$$\sigma = \sqrt{43.4}$$

$$\sigma = \underline{\underline{6.59}}$$

# 102 Variance and standard deviation of a Continuous Frequency Distribution

1) Calculate the variance and standard deviation of a continuous Frequency distribution

class interval 30-40 40-50 50-60 60-70 70-80 80-90 90-100

frequency 3 7 12 15 8 3 2

Using step deviation method.

We know that

$$\sigma = \sqrt{\frac{\sum f d^2}{\sum f} - \left(\frac{\sum f d}{\sum f}\right)^2 \times c}$$

Where  $d = \frac{x - A}{c}$

$c \rightarrow$  class interval

$A \rightarrow$  Assumed value (middle value)

| class interval | Frequency (fi) | mid value (xi) | $(\frac{x-A}{c})$ | $(\frac{x-A}{c})^2$ | $(\frac{x-A}{c}) \cdot d$ | $d^2$ | $f d^2$    | $f d$      |
|----------------|----------------|----------------|-------------------|---------------------|---------------------------|-------|------------|------------|
| 30-40          | 3              | 35             | -3                | 9                   | -3                        | 9     | 27         | -9         |
| 40-50          | 7              | 45             | -2                | 4                   | -2                        | 4     | 28         | -14        |
| 50-60          | 12             | 55             | -1                | 1                   | -1                        | 1     | 12         | -12        |
| 60-70          | 15             | A (65)         | 0                 | 0                   | 0                         | 0     | 0          | 0          |
| 70-80          | 8              | 75             | 1                 | 1                   | 1                         | 1     | 8          | 8          |
| 80-90          | 3              | 85             | 2                 | 4                   | 2                         | 4     | 12         | 6          |
| 90-100         | 2              | 95             | 3                 | 9                   | 3                         | 9     | 18         | 6          |
|                | <u>50</u>      |                |                   |                     |                           |       | <u>105</u> | <u>-15</u> |

We know that

$$\sigma = \sqrt{\frac{\sum fd^2}{\sum f} - \left(\frac{\sum fd}{\sum f}\right)^2 \times c}$$

$$\sigma = \sqrt{\frac{105}{50} - \left(\frac{15}{50}\right)^2 \times 10}$$

$$\sigma = \sqrt{2.1 - \frac{225}{2500} \times 10}$$

$$\sigma = \sqrt{2.1 - 0.09 \times 10}$$

$$\sigma = \sqrt{2.1 \times 100}$$

$$\sigma = \sqrt{20.1}$$

$$\sigma = \sqrt{21 - 0.09 \times 10}$$

$$\sigma = \sqrt{20.91 \times 10}$$

$$\sigma =$$

2) Calculate the variance and standard deviation of the continuous frequency distribution

|      |      |       |       |       |       |       |       |
|------|------|-------|-------|-------|-------|-------|-------|
| C.I: | 0-10 | 10-20 | 20-30 | 30-40 | 40-50 | 50-60 | 60-70 |
| F:   | 8    | 12    | 17    | 14    | 9     | 7     | 4     |

16.67

| class<br>Interval | Frequency<br>(f) | midvalue<br>(xi) | $x-A$ | d  | $d^2$ | $fd$       | $fd^2$     |
|-------------------|------------------|------------------|-------|----|-------|------------|------------|
| 0-10              | 8                | 5                | -30   | -6 | 36    | -48        | 288        |
| 10-20             | 12               | 15               | -20   | -4 | 16    | -48        | 192        |
| 20-30             | 17               | 25               | -10   | -2 | 4     | -34        | 68         |
| 30-40             | 14               | A (35)           | 0     | 0  | 0     | 0          | 0          |
| 40-50             | 9                | 45               | 10    | 2  | 4     | 18         | 36         |
| 50-60             | 7                | 55               | 20    | 4  | 16    | 28         | 112        |
| 60-70             | 4                | 65               | 30    | 6  | 36    | 24         | 144        |
|                   | <u>71</u>        |                  |       |    |       | <u>-57</u> | <u>840</u> |

$C=5$ ,  $d = \frac{x-A}{C}$

We know that step deviation method -

$$\sigma = \sqrt{\frac{\sum fd^2}{\sum f} - \left(\frac{\sum fd}{\sum f}\right)^2 \times C}$$

$$\sigma = \sqrt{\frac{840}{71} - \left(\frac{-57}{71}\right)^2 \times 5}$$

$$\sigma = \sqrt{11.83 - (-0.8)^2 \times 5}$$

$$\sigma = \sqrt{11.83 - 0.64 \times 5}$$

$$\sigma = \sqrt{11.19 \times 5}$$

$$\sigma = 3.345 \times 5$$

$$\sigma = \underline{\underline{16.7}}$$

## Coefficient of Variation

Coefficient of variation is defined as the Ratio of the standard deviation " $\sigma$ " to the arithmetic mean  $\bar{x}$  and it is often expressed as a percentage

$$\rightarrow \text{Coefficient of Variance} = \frac{\sigma}{\bar{x}} \times 100$$

1. The scores of two cricketers A and B in 10 Innings are given here. Find who is a better run getter and who is more consistent.

<sup>scores</sup>  
~~Score~~ of A    40   25   19   80   38   8   67   121   66   76  
 $\bar{x}$

<sup>scores</sup>  
~~Score~~ of B    28   70   31   0   14   111   68   31   25   4  
 $y_i$

For cricketer A

| $x$ | $d = x - \bar{x}$ | $d^2$ |
|-----|-------------------|-------|
| 40  | -14               | 196   |
| 25  | -29               | 341   |
| 19  | -35               | 1225  |
| 80  | 26                | 676   |
| 38  | -16               | 256   |
| 8   | -46               | 2116  |
| 67  | 13                | 169   |
| 121 | 67                | 4489  |
| 66  | 12                | 144   |
| 76  | 22                | 484   |
| 540 |                   | 10596 |

$$\bar{x} = \frac{\sum x}{n} = \frac{540}{10} = \underline{\underline{54}}$$

For cricketer B

$$d = y - \bar{y} \quad d^2$$

28

70

31

10

14

11

68

31

25

4

380

10674

$$\begin{aligned}\bar{y} &= \frac{\Sigma y}{n} \\ &= \frac{380}{10} \\ \bar{y} &= 38\end{aligned}$$

We know that

$$\text{Coefficient of variance of A} = \frac{\sigma_x}{\bar{x}} \times 100$$

$$\text{standard deviation} = \sigma_x = \sqrt{\frac{\Sigma d^2}{n}} \quad (\text{Actual mean method})$$

$$\sigma_x = \sqrt{\frac{10596}{10}} = \sqrt{1059.6} = 32.5$$

$$\begin{aligned}\text{Coefficient of variance of A} &= \frac{32.5}{54} \times 100 \\ &= \underline{60.28}\end{aligned}$$

$$\text{Coefficient of variance of B} = \frac{\sigma_y}{\bar{y}} \times 100$$

$$\text{standard deviation} = \sigma_y = \sqrt{\frac{\Sigma d^2}{n}} \quad (\text{Actual method})$$

$$\sigma_y = \sqrt{\frac{10674}{10}} = \sqrt{1067.4} = 32.67$$

$$\text{Coefficient of variance of B} = \frac{32.67}{38} \times 100 = \underline{86}$$

Since  $\bar{x} > \bar{y}$ ,  $A > B$

A is better than run getter

But coefficient variance of A < Coefficient of variance of B

$$\therefore 60.28 < 86$$

Note: When mean value is integer using actual method.

Q4/02

Skewness:

It measures the degree of asymmetry of a distribution.

→ Karl Pearson's Coefficient of skewness =

$$\frac{\text{mean} - \text{mode}}{SD} = \frac{\bar{x} - \text{mode}}{s}$$

Q3 Calculate the Karl Pearson's Coefficient of Skewness for the following data.

Variable 0-5 5-10 10-15 15-20 20-25 25-30 30-35 35-40

Frequency 2 5 7 13 21 16 8 3

To Find mean  $\bar{x} = A + \frac{\sum fd}{\sum f} \times c$

To Find standard deviation  $s = \sqrt{\frac{\sum fd^2}{\sum f} - \left(\frac{\sum fd}{n}\right)^2} \times c$

$$d = \frac{x - A}{c}$$

$$\text{To find mode} = 1 + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times c$$

| Variable | Frequency                     | mid Value<br>( $\frac{x+A}{2}$ ) | d  | fd        | d <sup>2</sup> | fd <sup>2</sup> |
|----------|-------------------------------|----------------------------------|----|-----------|----------------|-----------------|
| 0-5      | 2                             | 2.5                              | -4 | -8        | 16             | 32              |
| 5-10     | 5                             | 7.5                              | -3 | -15       | 9              | 45              |
| 10-15    | 7                             | 12.5                             | -2 | -14       | 4              | 28              |
| 15-20    | (13) <sup>f<sub>1</sub></sup> | 17.5                             | -1 | -13       | 1              | 13              |
| (20)-25  | (21) <sup>f</sup>             | (22.5) <sup>A</sup>              | 0  | 0         | 0              | 0               |
| 25-30    | (16) <sup>f<sub>2</sub></sup> | 27.5                             | 1  | 16        | 1              | 16              |
| 30-35    | 8                             | 32.5                             | 2  | 16        | 4              | 32              |
| 35-40    | 3                             | 37.5                             | 3  | 9         | 9              | 27              |
|          | <u>75</u>                     |                                  |    | <u>-9</u> |                | <u>193</u>      |

$$\text{mean} = \bar{x} = A + \frac{\sum fd}{\sum f} \times c$$

$$= 22.5 + \frac{-9}{75} \times 5$$

$$= 22.5 + (-0.12) \times 5$$

$$= 21.9$$

$$\text{Standard deviation } \sigma = \sqrt{\frac{193}{75} - \left(\frac{-9}{75}\right)^2 \times 5}$$

$$= \sqrt{2.573 - (-0.12)^2 \times 5}$$

$$= \sqrt{2.573 - 0.0144 \times 5}$$

$$= \sqrt{2.558 \times 5}$$

$$= 1.6 \times 5$$

$$= 8$$

$$\text{mode} = 1 + \frac{f_1 - f_2}{2f_1 - f_2} \times c$$

$$= 20 + \frac{21 - 13}{2(21) - 13 - 16} \times 5$$

$$= 20 + \frac{8}{42 - 29} \times 5$$

$$= 20 + \frac{8}{13} \times 5$$

$$= 20 + 0.615 \times 5$$

$$= \underline{\underline{23.08}}$$

We know that

Karl's Pearson's coefficient of skewness is

$$= \frac{\text{mean} - \text{mode}}{s}$$

$$= \frac{21.9 - 23.08}{8}$$

$$= \underline{\underline{-0.148}}$$

Kurtosis :

→ It measures the degree of peakedness of a distribution.

# Descriptive statistics

## Define statistics

Statistics is a tool in the hands of mankind to translate complex facts into simple and understandable statement facts.

Statistical methods

- collection of data
- organization of data
- presentation of data
- Analysis of data
- Interpretation of data

## \* Collection of data

- primary data
- Secondary data

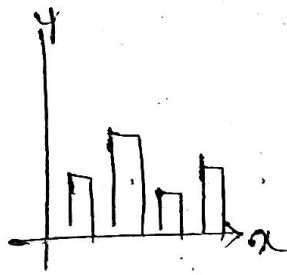
Population : large data

Sample : Small data

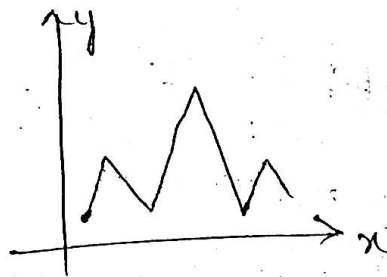
Data visualization :

- Bar chart
- line chart
- scatter plot
- spark line
- pie chart

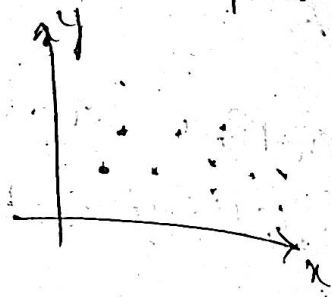
Bar chart



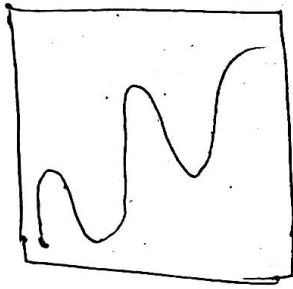
line chart



Scatter plot



Sparkline



• pie chart

